

Harsath Kaliappa Gounder Thangavel

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Education

York University

Bachelor of Science (BSc) in Computer Science (Honours)

Toronto, Canada

January 2020 – December 2024

Professional Experience

Software Engineer

Privex Inc.

October 2023 – January 2024

London, United Kingdom

- Developed an AI-driven support assistant using Retrieval-Augmented Generation (RAG) on a self-hosted LLM, reducing staff workload by handling common inquiries.
- Implemented REST APIs to interface with the LLM and integrated Discord APIs for customer queries and staff data updates.
- Built data ingestion and management workflows to process context data from website documentation, bulk files, support tickets, and Discord announcements.
- Built and maintained infrastructure using MySQL for chunked data storage and ChromaDB for embedding retrieval.
- Containerized the Flask API with Docker for development and production deployments.
- Created systemd service files and monitoring scripts for system reliability.
- Tech Stack:** Python (Flask, ChromaDB, Langchain), MySQL, Linux

Software Engineer

CogniPet AG

April 2019 – December 2019

Zürich, Switzerland

- Developed Pet Breed Detection system using Xception neural network and transfer learning, achieving 94% accuracy across 120 dog breeds.
- Implemented REST APIs for model inference and integrated them into CogniPet's Android and iOS apps using AWS EC2.
- Built data management workflows for user-uploaded pet images in AWS S3 with GDPR compliance.
- Explored video-based techniques for pet facial recognition through multi-frame feature extraction analysis.
- Tech Stack:** Python (Flask, Keras), AWS (EC2, S3)

Founder and Full-Stack Web Developer

TaffyHub.com

August 2017 – July 2019

Coimbatore, India

- Developed and launched school social networking platform, reaching 500+ registered users and 150+ daily active users.
- Implemented news feed, friend recommender, content sharing, engagement system, and premium membership features.
- Built backend with PHP and MySQL, frontend with JavaScript, jQuery, and Bootstrap, using Memcached for query optimization.
- Self-hosted the platform on Linux infrastructure, implementing reverse proxies, automated backups, and system maintenance as part of DevOps workflow.
- Generated \$100/month revenue through Google AdSense and premium memberships.
- Tech Stack:** PHP, JavaScript, MySQL, Memcached, Linux (rsync, cron)

Projects

Jix Interpreter | C Interpreter

- Developed an interpreter for a dynamically typed programming language.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented basic functional programming support, including first-class and higher-order functions.
- Tested using automated tests with CI/CD pipelines.
- Tech Stack:** C, CMake, GitHub Actions

Blueth | C++ Networking Library

- Developed a highly concurrent and event-driven networking framework for Linux in modern C++17.
- Implemented event-loop abstractions, HTTP parsers, SSL proxies, thread-pool executors, and codec parsers.
- Implemented thread-safe cache containers and synchronization mechanisms for multithreaded environments.
- Tested using Google Test framework and benchmarked through Google Benchmark library.
- Tech Stack:** C++17, Linux async I/O, CMake, GitHub Actions

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C++, C, SQL, PHP, Perl, Bash, HTML/CSS
Technologies: Linux, PostgreSQL, Docker, Kubernetes, MongoDB, Memcached, Redis, AWS, Neo4j
Frameworks: Node.js, React, Flask, Django, Langchain
Tools: Git, CI/CD, CMake, Maven, Linux Containers (LXC/LXD)